Low-Resource Legal Machine Translation:
A Nepali–English Case Study with Directional Analysis

[Authors]

Research Problem

- Legal translation for Nepali–English is severely under-resourced, lacking domain-specific parallel corpora
- Existing multilingual MT models are trained on general-domain data, performing poorly on legal texts requiring terminological precision and structural fidelity
- Unclear whether bi-directional training benefits or harms legal MT quality in extreme low-resource settings (¡6K sentence pairs)

Core Research Questions

1. Can multilingual pre-trained models achieve adequate legal translation quality with ¡6K high-fidelity parallel sentences?
2. Does bi-directional fine-tuning improve or degrade translation quality compared to uni-directional approaches in legal domain?
3. What linguistic error patterns persist across different model architectures (mBART vs. NLLB)?
4. How do directional asymmetries manifest in morphologically rich target languages?

Novel Contributions

- **Dataset:** First publicly available sentence-aligned Nepali–English corpus derived exclusively from authoritative legal documents (Constitution, Muluki Ain, statutes)—over 5,000 expert-validated pairs
- **Empirical Finding:** Bi-directional training causes substantial degradation in English→Nepali performance while maintaining Nepali→English quality—directional interference in legal MT
- **Methodological Insight:** Direction-specific fine-tuning outperforms joint training for legal translation in morphologically asymmetric language pairs under resource constraints
- **Error Taxonomy:** Systematic analysis of legal terminology, modality, and clause structure errors specific to Nepali legal MT

Methodology Overview

Corpus Construction

- Source documents: Constitution of Nepal, Muluki Ain, statutory laws, regulations
- OCR pipeline with Devanagari-optimized PyTesseract, extensive manual correction
- Semi-automatic sentence alignment with manual verification by law students
- Quality control: preservation of legal semantic equivalence, terminological consistency, syntactic completeness

Experimental Design

- Models: mBART50, NLLB-200 (multilingual pre-trained transformer architectures)
- Training configurations: (1) Uni-directional separate models, (2) Bi-directional joint model
- Identical train/dev/test splits and hyperparameters across all experiments
- Evaluation: BLEU, chrF++ (automatic); adequacy, fluency, legal fidelity (human)

Key Findings (Preliminary)

- Domain-adapted models significantly outperform baseline multilingual models on legal test sets
- **Directional asymmetry:** Bi-directional training degrades English→Nepali by 3–7 BLEU points while maintaining Nepali→English performance
- Error analysis reveals increased legal term mistranslation and modality errors under bi-directional training
- Uni-directional models show better preservation of complex clause structures in statutory language

Novelty Assessment for ICAIL

Strong Points

- First work addressing legal MT for Nepali–English with authentic legal corpus
- Counterintuitive finding: bi-directional training harms legal MT quality (challenges common multilingual MT practices)
- Direct relevance to AI & Law: implications for multilingual legal AI systems and access to justice
- Reproducible dataset release supports future legal NLP research

Areas to Strengthen

- Theoretical explanation of representational interference mechanisms
- Comparison with monolingual legal corpora augmentation strategies
- Human evaluation with practicing legal professionals (not just students)
- Analysis of specific legal document types (constitutional vs. statutory vs. regulatory)
- Discussion of deployment considerations for legal practice

Suggested Paper Structure for ICAIL

1. **Introduction** (1 page)
    - Legal translation gap for Nepali; access to justice motivation
    - Research gap: no domain-specific corpus, no directional analysis
    - Contributions summary
2. **Related Work** (1 page)
    - Legal MT (high-resource languages)
    - Low-resource MT with multilingual models
    - Nepali NLP and legal corpora
3. **Corpus Construction** (1.5 pages)
    - Source documents and legal coverage

- OCR pipeline and quality control
- Annotation protocol with legal expertise
- Dataset statistics and release information

4. **Experimental Setup** (1 page)
   - Model selection rationale
   - Training configurations (uni vs. bi-directional)
   - Evaluation metrics and human evaluation protocol
5. **Results** (1.5 pages)
   - Automatic metrics by direction and configuration
   - Human evaluation results
   - Directional asymmetry analysis
6. **Legal Error Analysis** (1.5 pages)
   - Terminology errors with examples
   - Modality and normative force preservation
   - Clause structure and interpretability
   - Case studies from different legal document types
7. **Discussion** (1 page)
   - Why bi-directional training fails for legal MT
   - Morphological and syntactic asymmetry hypothesis
   - Implications for legal AI deployment
   - Recommendations for practitioners
8. **Conclusion** (0.5 page)
   - Summary of contributions
   - Future work: corpus expansion, newer LLMs, document-level context

Additional Experiments to Consider

- Ablation study: effect of corpus size (train on 50%, 75%, 100% to show data efficiency)
- Zero-shot baseline comparison with GPT-4, Claude, or other LLMs
- Cross-domain evaluation: test on non-legal Nepali text to measure domain specificity
- Attention analysis: visualize where models focus in legal vs. general text
- Inter-annotator agreement scores for human evaluation

Timeline Recommendation

- Week 1–2: Finalize experiments, complete human evaluation with additional annotators
- Week 3–4: In-depth error analysis, attention visualization, case studies
- Week 5–6: Draft paper sections, create figures and tables
- Week 7: Internal review, revisions
- Week 8: Submission preparation, proofreading

Reference Papers

- https://aclanthology.org/2024.sigul-1.7.pdf
- https://arxiv.org/pdf/2212.09811
- https://openreview.net/pdf?id=LuXTBI6GSh
- https://www.isca-archive.org/sltu_2018/acharya18_sltu.pdf